ASSIGNMENT No. 1

Problem Statement: Write a program in C to check whether a given year is a Leap Year.

Objective:
1. To understand the use of conditional statements in C programming.
2. To implement logical conditions for solving real-world problems.
3. To develop problem-solving skills using decision control structures.

Theory:

1. Theory for conditional statements with syntax (if, if-else, if-else ladder) are used in C programmi
2. Theory for a Leap Year – year that contains 366 days instead of the usual 365 days.

Rules:
- A year is a leap year if it divisible by 400 or
- A year is a leap year if it is divisible by 4 but not divisible by 100.

3 **Algorithm**

Step 1: START.
Step 2: Declare an integer variable year
Step 3: Read the value of year.

Step 4: Check conditions:

- If year is divisible by 400, then it is a leap year.
- Else if year is divisible by 100 then it is not a leap year.
- Else if year is divisible by 4 then it is a Leap year.
- Else it is a not a leap year.

Step 5: Display whether the year is a leap year or not.

Step 6: STOP.

Platform: Linux.

Input:

```
# include <stdio.h>
int main() {
    int year;
    printf("Enter a year: ");
    Scanf("%d", &year);

    if (year % 400 == 0) {
        printf("%d is a Leap Year", year);
    }
    elseif (year % 100 == 0) {
    } printf("%d is Not a Leap Yr", year);
    else if (year % 4 == 0) {
        printf("%d is a Leap Yr", year);
    else {
        printf("%d is not a Leap yr", year)
    }
    return 0;
}
```

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Output:

Enter a year: 2024
2024 is a Leap year.

Sample Input
2024

Sample Output:
2024 is a Leap Year

Conclusion: Thus, the program successfully checks whether a given year is a Leap Year using conditional statements in C

FAQs:

1. What is Leap Year?
A. A leap year is a calender year that contains an additional day - February 29th - making the year 366 days long instead of the usual 365. This happens roughly every four years.

2. Why is the year 1900 not a leap year but 2000 is?
A. a) Year 1900 is divisible by 4 & is also divisible by 100
b) Year 1900 is not divisible by 400
c) It fails the final test for century years, so it not a leap year.

3) Which operators are used in this program?

A. a) Arithmetic Operators: The most important operator used is the Modulus operator (%)

b) Relational Operators: These are used to compare values & return a true or false result.

c) Logical Operators: To compine the three rules, logical operators are used to connect the conditions.

4) Difference between if & if-else statements.

A. a) The if statement is a single-path decision. It executes a specific block of code only if the condition is true. If the condition is true. If the condition is false, the program simply ignores that block & moves to the next part of the code

Ex: If you have enough money, buy a snack.

b) Then if-else statement: If-else statement is a two-path decision. It provides an alternative.

Ex: If it is a leap year, display "366 days", else, display "365 days".